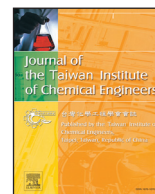




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Simultaneous removal of dye and heavy metal by banana peels derived hierarchically porous carbons

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ABSTRACT

The dye and heavy metal are two principal concomitant pollutants in industrial wastewaters, posing a serious threat to public health and the environment. Herein, we developed a novel strategy to convert banana peels into hierarchically porous carbon (BPCA) and porous carbon oxide (BPCAO) for the simultaneous removal of methylene blue (MB) and Co(II). The as-prepared carbons were systematically characterized by SEM, TEM, BET, FT-IR and XPS. Compared to BPCA, BPCAO showed a better adsorption performance due to its abundant surface oxygen-containing groups. In the mono-component systems, both MB and Co(II) adsorption onto BPCAO were well described by the pseudo-second-order and Langmuir models. The maximum adsorption capacities of MB and Co(II) were calculated to be 1303.54 and 122.39 mg/g at 298 K, respectively. In the binary system, the interaction between MB and Co(II) was determined and it depended on the initial concentration of Co(II). The adsorption mechanism of MB involved electrostatic interaction, π - π stacking and hydrogen-bonding while that of Co(II) was mainly ruled by the complexation and/or ion-exchange. This study indicated BPCAO could be a favorable biomass-derived adsorbent for the simultaneous removal of the dye and heavy metal.

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1. Introduction

The dye and heavy metal are two principal concomitant pollutants derived from many industries such as dyestuff, textile, leather, printing, paper and plastic [1–3], which constantly brings about a severe environmental issue due to their high toxicity for human health. Numerous works have been devoted to removing the single component (dyes or heavy metals) [4–7]. However, the impact of the co-contaminants in real sewages was always ignored, leading to the relatively low ecological remediation efficiency. To date, there are few attempts have been posed to explore the simultaneous removal of dyes and heavy metals [3,8]. Moreover, different physio-chemical properties of dyes and heavy metals make the disposal of co-polluted effluent more challenging [3]. Therefore, it is urgent to develop novel approaches to eliminate these harmful pollutants simultaneously.

According to the previous reports [1,4], the adsorption technique, possessing the potential to optimize the cost and

capacity, has been believed as an ideal approach. Various efficient adsorbents, including carbon-based materials [5,7,9–18], biomass or biomaterials [10,19–29], minerals [30–33], polymers [5,6,34] and metal-organic frameworks [35–38], have been widely employed to alleviate the ever-increasing reluctant effluent pollution. Among them, carbon-based nanomaterials gain extensive attention because of their high surface area, controlled pore size distribution, and manipulatable surface chemistry [9,10]. Especially, as a major class of carbon-based materials, the biomass-derived carbonaceous adsorbent is a promising adsorbing material for practical wastewater treatment due to its low-cost and easy-access. Recently, several different biomaterials such as fruit peels [21], cotton [39], chitin [4,20] and oiltea shell [40] were processed to be adsorbents for environmental remediation. Unfortunately, most of biomaterials are scarce and simultaneous removal property of bio-carbons still remains unknown.

Banana is a worldwide consumed fruit—the global production of bananas is about 10.6 million metric tons in 2015 [41]. Our consumption of bananas creates a large volume of banana peels (BPs) since the peel makes up about 40% of banana's weight. A great majority of these waste BPs is disposed in garbage dumps or landfills, where they release greenhouse gases and toxic gases as they decompose [42]. In the sense of environmental protection and

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sustainable resource utilization, BPs is an ideal low-cost precursor for value-added adsorbents due to their ultra-high content of carbohydrates. Several emerging BPs-derived carbon-based materials such as activated carbon [43], carbon foam [44], biochars [41] and pseudographite [42] have been evaluated as advanced adsorbents or battery anodes. In spite of these advances, no attempts have been undertaken to prepare BPs-derived carbonaceous sorbents for simultaneous removal of multiple pollutants. Further studies still need to be carried out to provide new insights into the surface chemistry of BP-derived carbons and simultaneous removal process.

In this study, we proposed a novel strategy to convert BPs into well-structured porous carbons (*i.e.*, BPCA and BPCAO) for the simultaneous removal of the dye and heavy metal. With the HNO_3 treatment of BPCA surface, BPCAO has abundant surface oxygen-containing groups. The adsorption affinity with the contaminants can be enhanced via the surface modification of BPCA. The physical-chemical structures of BPCA and BPCAO were characterized by several surface analysis approaches such as SEM, TEM, BET, FT-IR and XPS. The adsorption performance was systematically evaluated through the batch experiments. In order to reveal the behavior of BPCAO in the simultaneous removal of the dye and heavy metal, multi-component adsorption experiments were also carried out. Besides, the adsorption mechanism of BPCAO was further studied by BET, FT-IR and XPS measurements. To the best of our knowledge, this is the first time to prepare BPs-derived porous carbon for removing as-mentioned contaminants simultaneously. The present study also provides an alternative way both for value-added BPs disposal and efficient wastewater treatment.

2. Experimental

2.1. Materials and chemicals

Bananas were obtained from a fruit store nearby Zhejiang Sci-Tech University and produced in Hainan Province, China. Potassium hydroxide (KOH), nitric acid (HNO_3), methylene blue (MB), rhodamine B (RhB), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{CoCl}_2 \cdot 5\text{H}_2\text{O}$ were analytical grade and purchased from Aladdin. Deionized water was used in all the experiments.

2.2. Preparation of porous carbon adsorbents

The BPs was firstly washed several times with deionized water to remove undesired impurities. Then they were oven-dried at 60°C for 5 days and pulverized. After that, 100 g dried BPs was dispersed in 100 mL 8 M KOH solution to form a suspension, which was further treated with ultrasound for 1 h at the frequency of 40 KHz. The suspension was soaked for another 24 h under air conditions without stirring to guarantee the full interaction between KOH and carbon sources in BPs. The suspension was then centrifuged to collect treated BPs (a mixture of BPs/KOH). The supernatant (residual KOH solution) was also collected for further reuse. The obtained mixture was dried at 80°C for 24 h, receiving about 125 g dried BPs/KOH mixture. The dehydration of the mixture was achieved by thermal treatment at 400°C for 1 h in a muffle furnace under nitrogen conditions. Subsequently, the temperature was raised to 800°C with a heating rate of $5^\circ\text{C}/\text{min}$ and maintained for 1 h to activate the mixture. After cooling down under nitrogen, the sample was repeatedly washed with deionized water until the pH value reached 7. Finally, 11.2 g BPCA was obtained after drying at 80°C for 3 h.

3.0 g BPCA was dispersed in 100 mL 3 M HNO_3 and agitated for 3 h at 80°C under reflux. The excess HNO_3 solution, which could be reused, was removed by filtration and the sample was washed

to a neutral pH value with deionized water. Finally, 2.77 g BPCAO was received after drying the washed sample at 80°C for 3 h.

2.3. Characterizations

Scanning electron microscopy (SEM), transmission electron microscopy (TEM), N_2 adsorption-desorption isotherms, Fourier transform infrared (FT-IR) spectroscopy and X-ray photoelectron spectroscopy (XPS) were used to characterize the dried BPs and prepared adsorbents. SEM images were recorded on an emission scanning electron microscope (Ultra55, Zeiss, Germany). TEM images were obtained from JEM-2100 (JEOL, Japan) operated at an acceleration voltage of 200 KV. N_2 adsorption-desorption isotherms were performed by using a surface area and pore size analyzer (ASAP2020M, Micromeritics, USA) under 77 K. The specific surface area and pore size distribution of adsorbents were evaluated by the Brunauer–Emmett–Teller method. FT-IR spectra were measured with a Fourier transform infrared spectrometer (Vertex 70, Bruker, Germany) with the KBr disc method. XPS spectra were determined by an X-ray photoelectron spectrometer (K-Alpha, Thermo, USA) using $\text{Al-K}\alpha$ ($h\nu = 1486.71\text{ eV}$) as a radiation source. The zeta-potentials (pH_z) of the BPCA and BPCAO were measured using NanoSizer Nano-ZS90 (Nano ZS90, Malvern, United Kingdom) by loading 0.08 wt% suspension into the instrument.

2.4. Adsorption experiments

In order to assess the adsorption performance of resulting adsorbents, methylene blue (MB) and rhodamine B (RB) were selected to be the model dyes while Cu(II) and Co(II) were used as model heavy metals in the batch experiments. The stock solution of each dye (2000 mg/L) and heavy metal ion (1000 mg/L) were prepared by dissolving a certain amount of chemicals in deionized water and diluted to the desired concentrations for adsorption experiments. The pH of the prepared solution was not adjusted, making it more appropriate for practical sewage treatment. After that, 20 mg adsorbent was added to the diluted solution (25 mL) to form an operating mixture, which was further processed under controlled temperature (25°C) with agitation (200 rpm). After agitating for 24 h, the supernatant was received from the suspension after filtration using a $0.45\text{ }\mu\text{m}$ membrane. Uv-Vis spectroscopy (Lambda 35, Elmer Perkin, USA) was used to determine the concentrations of dye solution by measuring the absorbance at the maximum absorption wavelength. The concentrations of heavy metals in solution were analyzed via atomic absorption spectrometer equipped with a graphite furnace (Sollar M6, Thermo, USA). All the adsorption experiments were carried out in triplicate, and the performance of adsorption was evaluated by the equilibrium adsorbed capacity (Q_e) of pollutants on adsorbents and the distribution coefficient (K_d), describing as Eqs. (1) and (2), respectively.

$$Q_e = \frac{(C_0 - C_e)V}{M} \quad (1)$$

$$K_d = \frac{C_0 - C_e}{C_e} \times \frac{V}{M} \quad (2)$$

where C_0 (mg/L) and C_e (mg/L) are the initial concentration and the equilibrium concentration, respectively. V (mL) is the volume of suspension. M (mg) is the mass of the adsorbent.

For the adsorption kinetic studies, the concentrations of dye and heavy metal ion were 800 mg/L and 100 mg/L, respectively. To obtain the adsorption isotherms, 20 mg adsorbent was severally added into a set of 25 mL solutions with a series concentration of adsorbates and incubated for 24 h at 298 K, 308 K and 318 K, respectively.

Table 1

Textual properties of BPCA and BPCAO before and after pollutants adsorption.

Sample	BET-surface area (m ² g ⁻¹)	Total pore volume (cm ³ g ⁻¹)
BPCA	1396.60	0.75
BPCAO	1238.47	0.64
BPCAO + MB	566.17	0.32
BPCAO + Co(II)	1029.25	0.57

2.5. Multi-component adsorption experiments

Multi-component adsorption experiments were performed with the binary systems of Co(II)-MB on BPCAO. In Co(II)-MB system, batch experiments were implemented using an initial concentration ranging from 10 to 120 mg/L and 200 to 1000 mg/L for Co(II) and dyes, respectively. Especially, a full factorial design (Table S1), in which the main factors were the initial concentrations of each pollutant, was introduced to these experiments. The pH and contact time in binary batch experiments was about 7.0 (without any adjustment) and 24 h, respectively. The matching adsorption capacities of each component were selected to be the response variables of this factorial design. In order to make further comparisons, the adsorption capacities of single-component adsorption experiments at the same batch conditions as binary ones were also obtained.

3. Results and discussion

3.1. Characterizations of adsorbents

The morphologies and microstructures of BP, BPCA and BPCAO are shown in SEM and TEM images (Fig. 1). Fig. 1A and B (SEM images of BP) evidently show that the crushed pulverized BP is a smooth bulk with a particle size of about 20 μm. Fig. 1C and D (SEM images of BPCA) demonstrate that BPCA has a sponge-like porous microstructure, leading to a high specific surface and the rapid transformation of pollutants from outside surface into the inner porous network. SEM images of BPCAO (Fig. 1E and F) imply that the main structure of porous carbons is maintained after treatment with HNO₃. As a result, this structure can endow BPCA and BPCAO a strong pollutant removal performance. Fig. 1G and H (TEM images of BPCA) also show a strong evidence of porous morphology of BPCA. It's easy to find that numerous pores crowd tightly on the carbon sheet. Besides, TEM images of BPCAO (Fig. 1I and J) further indicate that the morphology and microstructure of BPCA are still remained after treating with HNO₃.

The N₂ adsorption-desorption isotherms of the BPCA and BPCAO are shown in Fig. 2A. Detailed information about their textual properties is listed in Table 1. BPCA and BPCAO possess a similar specific surface area, reaching up to 1396.60 and 1238.47 m²/g, respectively. This suggests that the microstructure of BPCA is maintained after HNO₃ treatment. Such high specific surface areas of the BPCA and BPCAO endows them with high pollutant adsorption capacities. The size distributions of the BPCA and BPCAO are shown in Fig. 2B. It can be obviously found that both of these two adsorbents possess a large amount of micropores and mesoporous in their structures, which may act as "adsorbed field" for pollutants.

The FT-IR spectra of prepared sorbents are displayed in Fig. 2C. The characteristic absorption peak of functional groups –OH (3390 cm⁻¹), C=O (1567 cm⁻¹) and C–O (1106 cm⁻¹) are observed in the BPCAO spectrum [39,45]. Whereas, no significant characteristic peaks are detected in the BPCA spectrum. These observations suggest that the BPCA is a carbonaceous adsorbent with few oxygen functional groups, and many oxygen functional groups (i.e., –OH, –COOH) are formed on the BPCAO surface after HNO₃ treatment. In addition, these groups can obviously change the

surface properties of BPCAO and enhance the pollutant removal of BPCAO by acting as binding sites.

Surface states of the BPCA and BPCAO are further analyzed using the XPS technique (Fig. 2D). In the spectrum of BPCA, the strong C 1s peak and weak O 1s peak reveal that the main element is carbon, which is consistent with the FT-IR spectrum of BPCA. In the spectrum of BPCAO, the stronger O 1s peak indicates that more oxygen functional groups are generated on the BPCAO surface by comparison to BPCA. Fig. 2E and 2F illustrate the high-resolution C 1s and O 1s XPS spectra of BPCAO, respectively. Since the binding energy of the sp²-hybridized graphite-like carbon atoms (C=C) and the sp³-hybridized carbon atoms (C–C) are 284.6 eV and 285.1 eV [39], one can verify that the skeleton of BPCAO is the same as BPCA. Besides, the C 1s XPS spectrum exhibits two additional peaks at 286.5 eV and 288.4 eV, which are attributed to C–O and C=O respectively [5,8,41,46,47]. Moreover, the O 1s XPS spectrum of BPCAO reveals two distinct peaks at 532.2 eV and 533.3 eV, which are allocated to the C–O groups and C=O groups, respectively [5,48].

Fig. 3A displays the result of the surface charge of BPCA and BPCAO in the solution pH range of 1.0–7.0. One can conclude that the isoelectric point (pH_{isp}) values of BPCA and BPCAO are about 3.66 and 1.92, respectively. Notably, the surface charge of solid is positive at pH < pH_{isp}, while a negative surface charge of solid is realized at pH > pH_{isp}. Therefore, BPCA and BPCAO possess negative charges in the deionized water (pH ≈ 7.0), providing electrostatic adsorption sites for positive pollutants (i.e., cationic dyes and metals). Specially, BPCAO possesses a more negatively charged surface in the deionized water than that of BPCA, implying a better adsorption capacity.

3.2. Adsorption studies

The adsorption capacities of BP, BPCA and BPCAO toward different pollutants are presented in Fig. 3B. It can be clearly seen that BP hardly adsorbs these contaminants (31.26 mg/g for MB, 21.72 mg/g for RB, 5.17 mg/g for Cu(II) and 4.23 mg/g for Co(II)). By comparison, BP-derived carbons (i.e., BPCA and BPCAO) exhibit high adsorption capacities to dyes and heavy metals. The adsorption capacities of BPCA toward MB, RB, Cu(II) and Co(II) are 744.39 mg/g, 520.29 mg/g, 62.51 mg/g and 80.89 mg/g, respectively. As for BPCAO, its adsorption capacities toward MB, RB, Cu(II) and Co(II) are 986.89 mg/g, 716.19 mg/g, 93.34 mg/g and 121.12 mg/g, respectively, demonstrating the enhancement of adsorption performance by the oxidation treatment. Furthermore, both of the BPCA and BPCAO possess higher adsorption capacities toward MB and Co (II) than BP. Particularly, compared to BPCA, BPCAO can adsorb more than 1.3 times and 1.5 times of MB and Co(II), respectively. This may be due to the generation of large amounts of oxygen functional groups on its surface. To better describe the adsorption process, the kinetics, isotherms and thermodynamics of MB and Co(II) adsorption on BPCAO were fully investigated.

3.2.1. Adsorption kinetics

Fig. 4A and B displays the effect of contact time on the adsorption of monocomponent contaminants onto BPCAO. It is found that the dyes and metals quickly adsorbed on BPCAO within the first 60 min, and accomplished their maximum Q_e at 240 min and 180 min, respectively. Subsequently, in order to further reveal the interaction mechanism of adsorption processes, the pseudo-first-order and pseudo-second-order kinetics were adopted to analyze the experimental data [49]. The two models are expressed as Eqs. (3) and (4):

$$\ln(Q_e - Q_t) = \ln Q_e - k_1 t \quad (3)$$

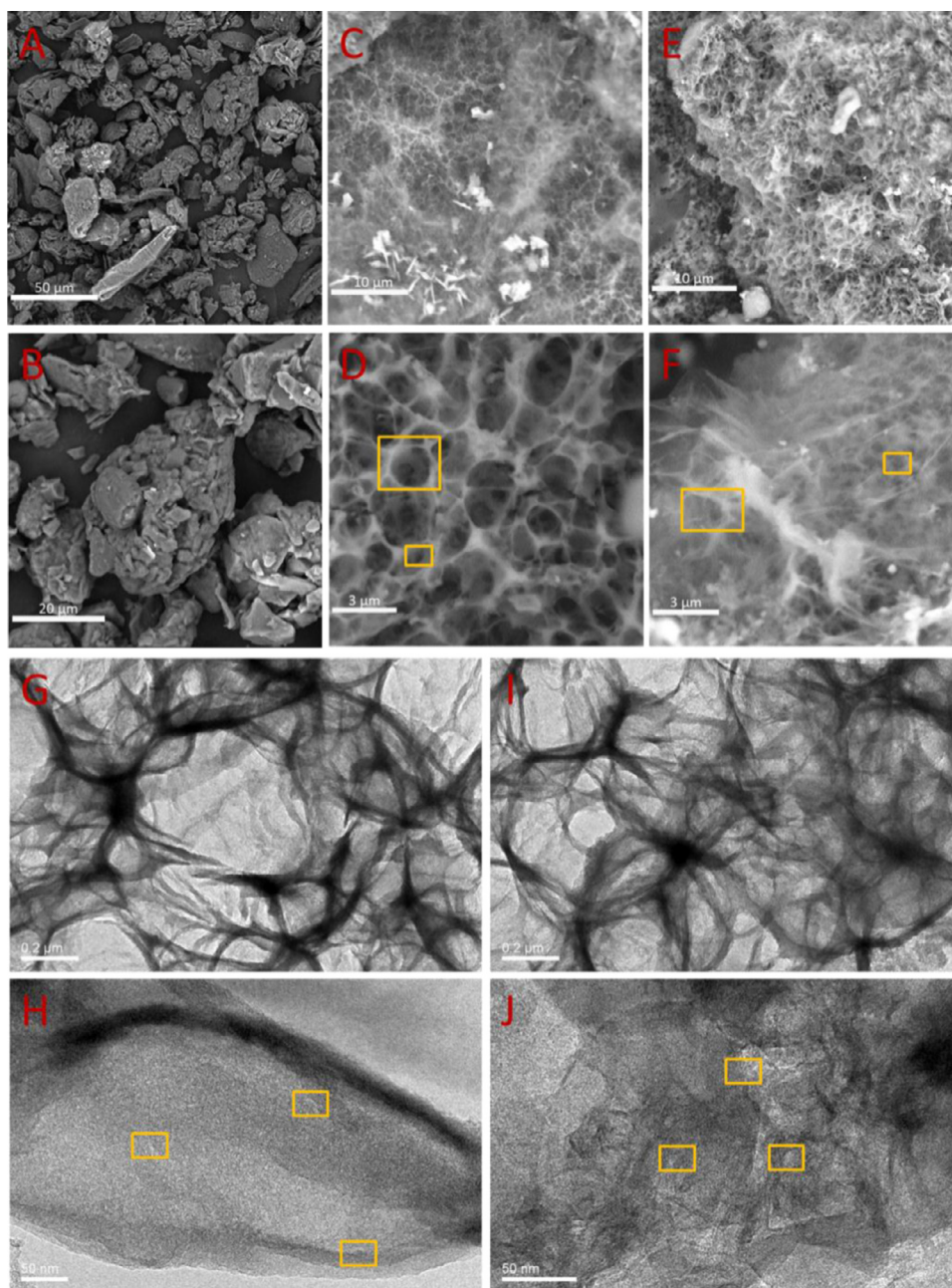


Fig. 1. SEM images of BP (A and B), BPCA (C and D) and BPCAO (E and F); TEM images of BPCA (G and H) and BPCAO (I and J).

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{1}{Q_e} t \quad (4)$$

where k_1 (min^{-1}) and k_2 ($\text{g mg}^{-1} \text{min}^{-1}$) represent the kinetic adsorption rate constants of the pseudo-first-order and pseudo-second-order models, Q_t is the adsorption capacity at contact time t (min), respectively.

The kinetic parameters such as the correlation coefficient R^2 and calculated Q_e , which are used to evaluate the validity of fitted models, are determined from the fitting results and shown in Table 2. The pseudo-second-order model is more suitable for simulating the adsorption kinetics of BPCAO due to its high the correlation coefficient (R^2), indicating that the interaction between adsorbates and adsorbents is mainly ruled by a chemical process [45]. A similar result was also reported by Long et al. [50]. In addition, from the view of adsorption rates, one can be

easily observed that all the k_2 s of pollutant adsorption on BPCAO ($k_2^{\text{MB}} = 1.401 \times 10^{-4}$, $k_2^{\text{Co(II)}} = 4.631 \times 10^{-4}$) are much higher than that of pollutant adsorption on BPCA ($k_2^{\text{MB}} = 7.748 \times 10^{-5}$, $k_2^{\text{Co(II)}} = 3.816 \times 10^{-4}$), which means the adsorption on BPCAO is quicker than the one on BPCA. This result can be clarified by the formation of a large amount of the oxygen-containing functional groups on BPCAO [39].

3.2.2. Adsorption isotherms

The adsorption isotherms of MB and Co(II) on BPCAO at temperatures of 298 K, 308 K and 318 K are given in Fig. 4C and D. It can be obviously seen that all the adsorption isotherms exhibit an increasing trend as C_e increases, and the increase gradually slows down as C_0 increases. Besides, MB and Co(II) adsorption on BPCAO increases with temperature increasing, indicating that a higher temperature can enhance pollutant adsorptions. Moreover,

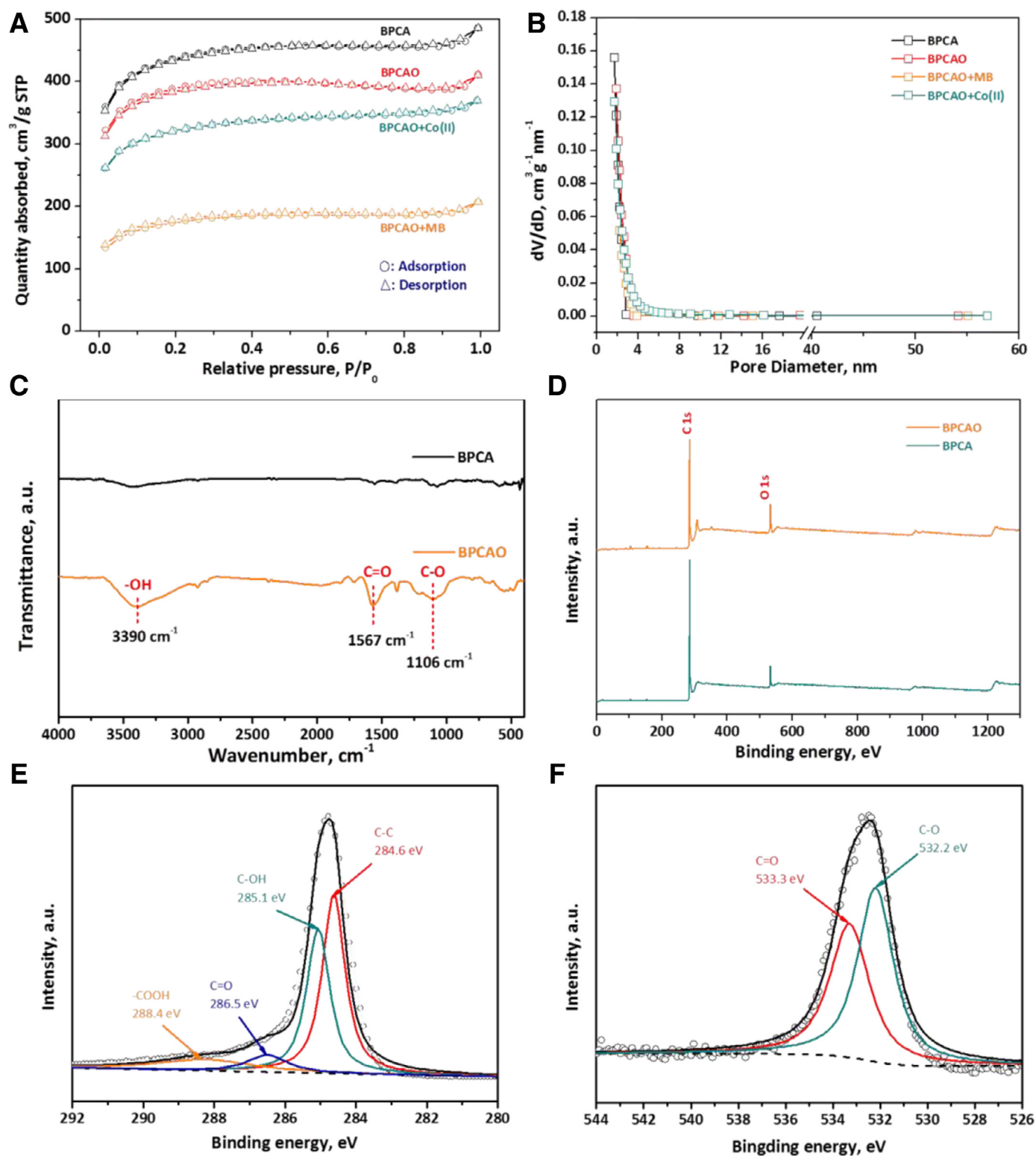


Fig. 2. BET specific surface areas of the BPCA and BPCAO before and after pollutants adsorption (A); size distribution of BPCAO before and after pollutants adsorption (B); FT-IR (C) and XPS (D) spectra of the BPCA and BPCAO; highly resolution XPS C1s spectrum (E) and O 1s spectrum (F) of BPCAO.

the two typical isotherms, i.e., Langmuir and Freundlich models, were utilized to fit the experimental results. The Langmuir and Freundlich isotherm equations are given as follows:[49]

$$Q_e = \frac{K_L Q_m C_e}{1 + K_L C_e} \quad (5)$$

$$Q_e = K_F C_e^n \quad (6)$$

where Q_m (mg/g), K_L (L/mg), K_F ($\text{mg}^{1-n} \text{L}^n \text{g}^{-1}$) and n are the maximum theoretical adsorption capacity, the Langmuir adsorption constant, the Freundlich equilibrium constant and the Freundlich exponent, respectively.

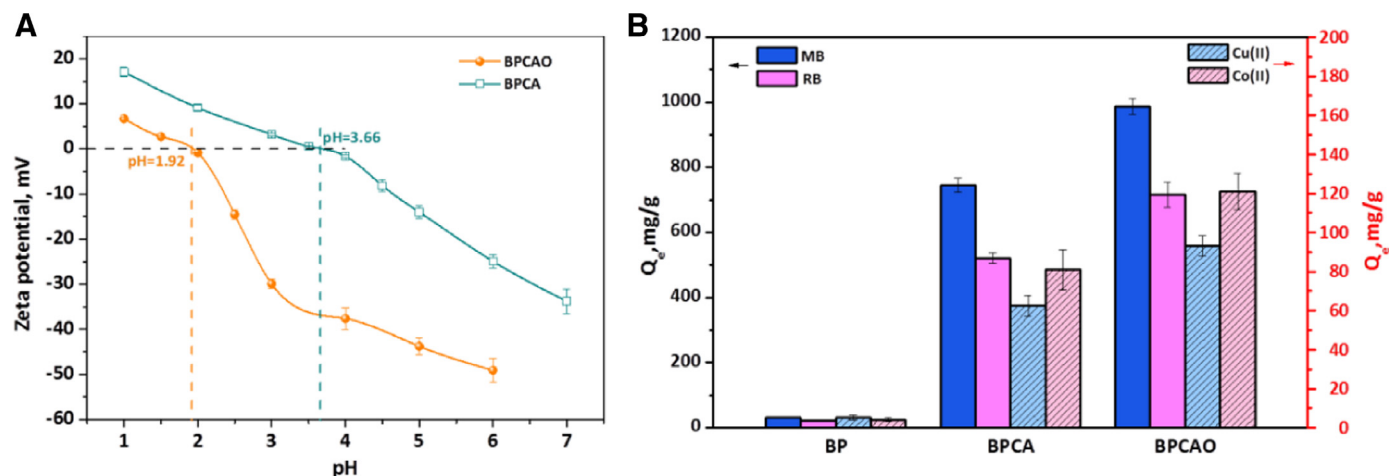


Fig. 3. Zeta potentials of BPCA and BPCAO at the pH range of 1.0 to 7.0 (A); comparison of adsorption capacities toward dyes and metals of BP, BPCA and BPCAO ($C_{0\text{ dye}} = 800 \text{ mg/L}$, $C_{0\text{ metal}} = 100 \text{ mg/L}$, $m = 20 \text{ mg}$, $V = 25 \text{ mL}$, $T = 298 \text{ K}$, $t = 24 \text{ h}$) (B).

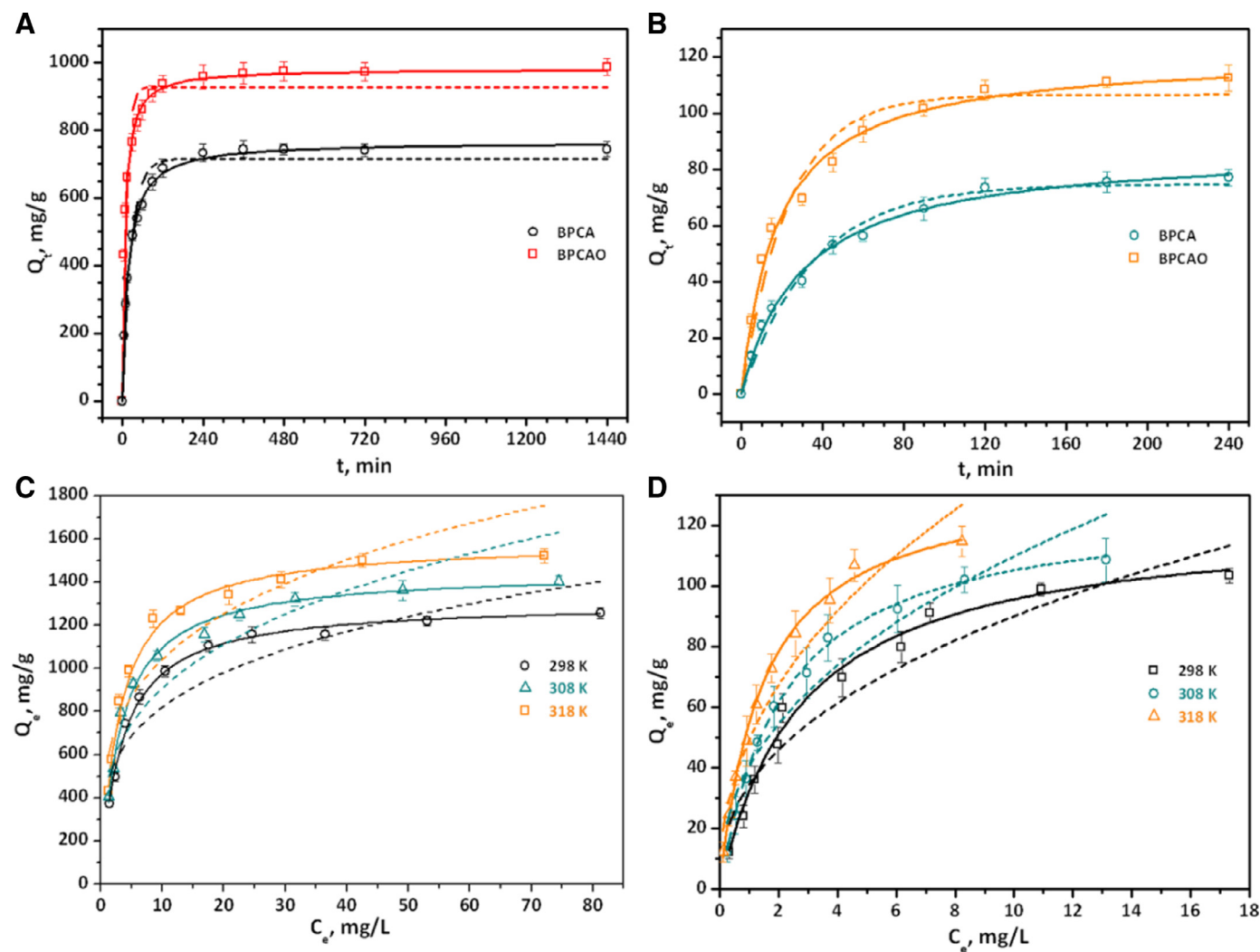


Fig. 4. Adsorption Kinetics of (A) MB and (B) Co(II) onto BPCA and BPCAO (Dashed line: pseudo-first-order fitting; Solid line: pseudo-second-order fitting); adsorption isotherms of (C) MB and (D) Co(II) onto BPCAO (Solid line: Langmuir model fitting; Dashed line: Freundlich model fitting).

Table 2
Kinetic parameters for MB and Co(II) adsorption on BPCAO.

Pollutants Adsorption	Pseudo-first-order			Pseudo-second-order		
	k_1 (min ⁻¹)	Q_e (mg/g)	R^2	k_2 (g mg ⁻¹ min ⁻¹)	Q_e (mg g ⁻¹)	R^2
BPCA + MB	0.0394	715.51	0.9662	7.748×10^{-5}	766.72	0.9972
BPCA + Co(II)	0.0285	74.64	0.9814	3.816×10^{-4}	87.76	0.9942
BPCAO + MB	0.0893	927.84	0.9548	1.401×10^{-4}	981.71	0.9970
BPCAO + Co(II)	0.0434	106.54	0.9647	4.631×10^{-4}	121.02	0.9918

Table 3
Langmuir and Freundlich model parameters of MB and Co(II) onto BPCAO.

Adsorbates	T (K)	Langmuir model			Freundlich model		
		Q_m (mg/g)	K_L (L/mg)	R^2	K_F (mg/g)(L/mg) ^{1/n}	n	R^2
MB	298	1303.53	0.2969	0.9948	464.16	0.2561	0.8714
	308	1447.44	0.3120	0.9920	455.38	0.2916	0.8710
	318	1584.95	0.3271	0.9934	588.22	0.2637	0.8687
Co(II)	298	122.39	0.3574	0.9948	34.36	0.4185	0.9406
	308	127.63	0.4675	0.9988	40.38	0.4344	0.9471
	318	135.78	0.6814	0.9971	49.35	0.4474	0.9575

Table 4
Thermodynamic parameters for MB and Co(II) adsorption on BPCAO.

Adsorbates	T (K)	ΔG (KJ/mol)	ΔH (kJ/mol)	ΔS (J mol ⁻¹ K ⁻¹)
MB	298	-13.62	14.28	93.68
	308	-14.62		
	318	-15.49		
Co(II)	298	-8.94	35.06	147.31
	308	-10.10		
	318	-11.90		

The non-linear fitted parameters calculated from these two isotherm models are listed in Table 3. As shown in Table 3, the R^2 of the Langmuir model is larger than that of the Freundlich model, and it is almost close to the unit one. Besides, fitting curves of Langmuir are better fitted to the experimental data. Based on the above analysis, the Langmuir model shows a better simulated result than the Freundlich model toward MB and Co(II) sorption on BPCAO, suggesting that the adsorption of MB and Co(II) are both a single-layer pattern [51].

3.2.3. Adsorption thermodynamics

To gain better insight into adsorption processes, the thermodynamic parameters of these processes (i.e., ΔG , standard Gibbs free energy change; ΔH , standard enthalpy change; ΔS standard entropy change) are calculated from the adsorption isotherms, which are collected at given temperatures. The specific calculated equations are expressed as:

$$\Delta G = -RT \ln K \quad (7)$$

$$\ln K = \frac{\Delta S}{R} - \frac{\Delta H}{RT} \quad (8)$$

where K is the thermodynamic equilibrium constant, and the values of $\ln K$ are calculated from plotting of $\ln K_d$ vs C_e and extrapolating C_e to zero (Fig. 6A) [49]; R represents the ideal gas constant (8.314 J mol⁻¹ K⁻¹); T (K) is the Kelvin temperature. The thermodynamic parameters are summarized in Table 4. From Table 4, one can deduce that pollutant adsorption on BPCAO is spontaneous processes because of the negative ΔG . The positive ΔH values reveal that pollutants sorption onto BPCAO are endothermic processes, which are well matched with the finding that a higher temperature can promote the adsorption [36,52]. This result may be derived from the fact that the complexation between pollutant and adsorbent is changed at different temperatures. The positive values

of ΔS indicate an increase of randomness at the solid/solution interface in the adsorption process [152,53]. This may be ascribed to the formation of sorbate/sorbent composites [39].

3.2.4. Simultaneous adsorption studies of multi-component system

The result of simultaneous adsorption of metals and dyes on BPCAO as well as the corresponding comparative consequence of the single-component system are both depicted in Fig. 5. In order to clearly uncover the interaction between Co(II) and dyes in the simultaneous adsorption process, the ratio of adsorption capacities (R_Q) is used to analyze the adsorption data. Its equation is described as:

$$R_Q = \frac{Q_{m,j}}{Q_{s,j}} \quad (9)$$

where $Q_{s,j}$ (mg/g) and $Q_{m,j}$ (mg/g) are the adsorbed amount of contaminant j in the single-component system and multi-component system with the equal concentration, respectively. The interaction can be verified as following three types according to the R_Q values [8,54]: (i) when $R_Q < 1$, antagonism, i.e., the presence of concomitant pollutant inhibits the adsorption of contaminant j ; (ii) when $R_Q = 1$, noninteraction, i.e., both contaminants do not affect each other; (iii) when $R_Q > 1$, synergism, i.e., the presence of concomitant pollutant enhances the adsorption of contaminant j . Fig. 5 shows the results of Q_e (Fig. 5A and 5C) and R_Q (Fig. 5B and 5D) values at different initial concentrations in the multi-component system. Basically, the adsorption of Co(II) is almost unchanged ($R_{Q, Co(II)} \approx 1.0$) at its low initial concentration (i.e., 10, 20, 40, 60 mg/g) in the presence of MB. Interestingly, strongly enhanced ($R_{Q, Co(II)} > 1.0$) sorption capacity of Co(II) is observed by the presence of MB at high Co(II) concentrations (i.e., 80, 100, 120 mg/g). Thus, the presence of MB does not pose a negative impact to the Co(II) adsorption, but indeed positively enhances the Co(II) adsorption at elevated Co(II) concentrations. Specifically, the adsorption capacities of Co(II) at the initial concentration of 200 mg/L in the monometal system ($Q_{e,[120,0]}$) is 105.63 mg/g, whereas 115.93 mg/g in Co(II)–MB binary system ($Q_{e,[120,1000]}$). These findings reveal an interactive effect between Co(II) and MB. This effect can be tentatively elucidated as following: on the one hand, based on the BET measurement and FT-IR spectra of BPCAO after pollutant adsorption (which will be discussed further in 3.2.5 Adsorption mechanism), one can deduce that there may be sufficient sorption sites both for Co(II) and MB at a relatively low Co(II) concentration. On the other hand, the sorbed MB will introduce nitrogen-containing groups that have been reported to be able to

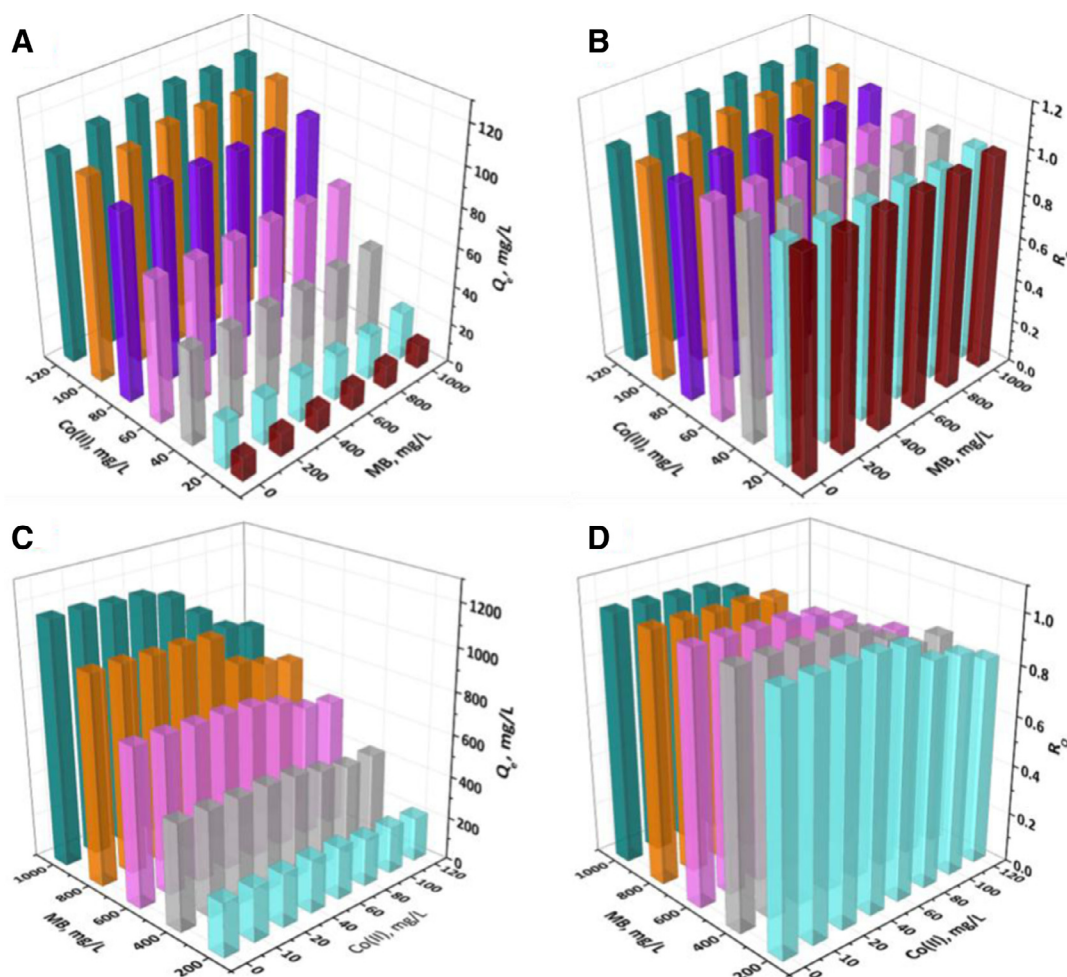


Fig. 5. Q_e and R_d values versus the initial concentrations of multi-component system for the simultaneous sorption of MB (A, B) and Co(II) (C, D).

complex metals [1], providing extra new sites for Co(II) adsorption, thus resulting in an enhanced adsorption capacity of Co(II) at a high Co(II) concentration.

Interestingly, the MB adsorption ($R_{Q,MB} \approx 1.0$) onto BPCAO is also unaffected by Co(II) at low initial Co(II) concentrations. However, in contrast to the Co(II) adsorption, significantly reduced sorption of MB ($R_{Q,MB} < 1.0$) occurs under comparatively high Co(II) concentrations (i.e., 80, 100, 120 mg/g), demonstrating a negative impact of Co(II) toward MB adsorption in the binary system. Besides, the higher MB concentration, the stronger negative effect it provides. This suggests more MB molecules are kept away from their original sorption sites. In other words, almost available sites are occupied by Co(II). These results may be derived from following reasons: (1) competitive adsorption between positively charged MB and positively charged Co(II) [54]; (2) steric hindering effect of the pore surrounding high-density complex between Co(II) and oxygen-containing groups (i.e., -OH and -COOH/-COO⁻) at high Co(II) concentrations.

3.2.5. Adsorption mechanisms

Generally, the adsorption may lead to a change of textual properties of the adsorbent. Therefore, N₂ adsorption-desorption measurements were carried out to clarify the textual change of BPCAO before and after pollutant adsorption. As shown in Fig. 2A and Table 1, the surface area and total pore volume of BPCAO after MB sorption dramatically decreases from 1238.47 to 566.17 m² g⁻¹ and from 0.64 to 0.32 cm³ g⁻¹, respectively. However, textual properties of BPCAO after Co(II) sorption are almost unchanged.

This indicates that MB with large molecular size could occupy the pore after adsorption while the pore could hardly be occupied by Co(II) due to its ultra small size. Besides, pore size distributions of BPCAO + MB and BPCAO + Co(II) (Fig. 2B) further support the finding, it is also in good accordance with the results of previous reports [39,47,55]. And this different textual changes in BPCAO may illustrate the simultaneous adsorption results at different Co(II) initial concentrations.

FT-IR technique was carried out to uncover the adsorption mechanisms. The FT-IR spectra of MB, BPCAO before and after pollutants adsorption are shown in Fig. 6B. The specific chemical groups of BPCAO have already been analyzed above. As for the FT-IR spectrum of MB, the peaks at 3413 cm⁻¹, 1594 cm⁻¹, 1493 cm⁻¹, 1389 cm⁻¹, 1326 cm⁻¹, 1245 cm⁻¹, 1170 cm⁻¹, 1136 cm⁻¹ and 882 cm⁻¹ are ascribed to the -OH of chemisorbed water molecule, C=C (C=N) bonds in the heterocycle of MB, deformation vibration of CH₂ in the benzene ring, CH₃ deformation vibration, benzene connected C-N stretch vibration, Ar-N deformation vibration, C=S stretch vibration, C-S stretch vibration and the wagging vibration of C-H in the aromatic ring of MB, respectively [47,56]. It is evidently clear that the FT-IR spectrum of BPCAO after MB sorption (BPCAO + MB) reveals several obvious changes. The peaks assigned to C=C (C=N), CH₂ in the benzene ring and benzene connected C-N stretch vibration shift from 1594 cm⁻¹ to 1629 cm⁻¹, 1493 cm⁻¹ to 1453 cm⁻¹, 1389 cm⁻¹ to 1372 cm⁻¹, respectively. These shifts are due to the attachment of MB on the BPCAO surface by π - π stacking interactions between the MB planar molecule and the BPCAO's hexagonal skeleton [47,57]. The shift

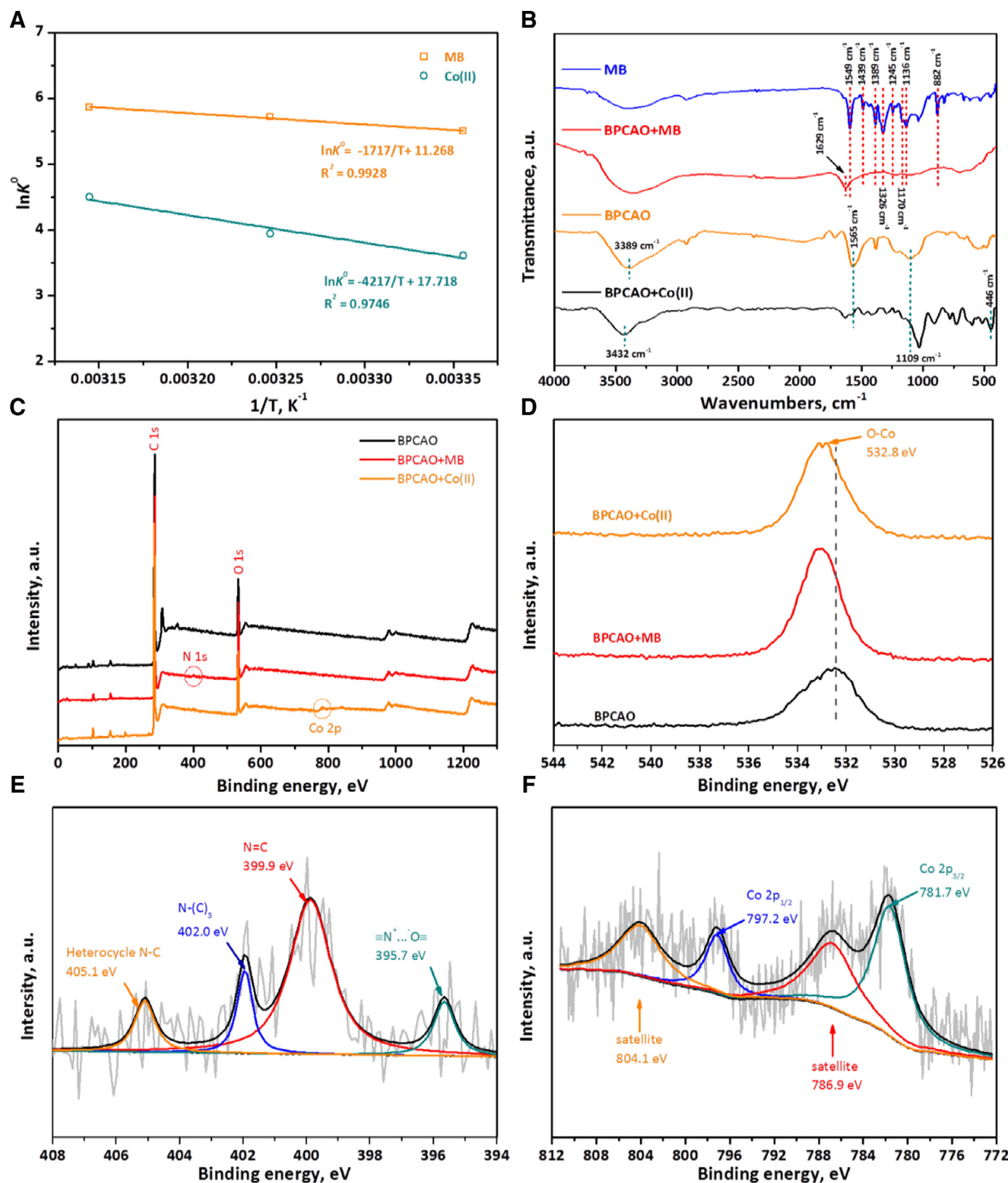


Fig. 6. The linear plot of $\ln K^0$ vs $1/T$ for MB and Co(II) adsorption on BPCAO at 298, 308 and 318 K (A); FT-IR spectra of MB, BPCAO before and after adsorption (B); Total scans of XPS spectra (C) and the high-resolution O 1s spectra (D) of BPCAO, BPCAO+MB and BPCAO+Co(II); the high-resolution N 1s spectrum (E) of BPCAO+MB; the high-resolution Co 2p spectrum (F) of BPCAO+Co(II).

(882 cm^{-1} to 806 cm^{-1}) of the aromatic C–H wagging vibration of MB further confirms the π - π stacking interactions. Additionally, the peaks corresponded to N-containing (C–N, Ar–N) and S-containing groups (C=S, C–S) have changed in respect of their location and intensity. This indicates that the N or S atom in N or S-containing groups may serve as the hydrogen-bonding acceptor for generating hydrogen bond between the hydroxyl group on the BPCAO surface [56]. In the Co(II) adsorption case (BPCAO + Co(II)), the –OH peak is broadened with a decrease intensity and it shifts from 3389 cm^{-1} to 3432 cm^{-1} after Co(II) adsorption. This suggests that the interaction between BPCAO and Co(II) may involve the ion exchange of –OH (C–OH/COOH) and Co(II). Moreover, the changes of C=O and C–O demonstrate that the complexation between Co(II) and C=O/C–O also participates the Co(II) adsorption process. Meanwhile, the peak at 446 cm^{-1} of Co–O is obviously observed at the spectra of BPCAO + Co(II) [58], further verifying the interaction of Co(II) and oxygen-containing functional groups.

XPS measurement was conducted to further verify the adsorption mechanisms. Fig. 6C shows XPS survey spectra of BPCAO before and after the adsorption of MB and Co(II). The peaks of N 1s and Co 2p are observed after adsorption, suggesting that the MB and Co(II) were adsorbed onto BPCAO. The high-resolution O 1s spectra (Fig. 6D) absolutely reveals that the oxygen-containing groups are involved in the sorption of MB and Co(II) since remarkable binding energy shifts of O 1s peaks are observed. Particularly, the peak at 532.8 eV in the O 1s spectrum of BPCAO + Co(II) implies the presence of O–Co in the Co(II) sorbed BPCAO [59], suggesting the interaction between O and Co via the complexation or/and ion exchange. This result also matches well with the FT-IR spectra. Additionally, the peak centered at 781.2 eV is assigned to Co 2p_{3/2} with a shakeup satellite at 786.9 eV and the peak centered at 797.2 eV is ascribed to Co 2p_{1/2} with a shakeup satellite at 804.1 eV (Fig. 6E). These two relatively stronger satellite peaks accompanied with 2p_{3/2} and 2p_{1/2} in BPCAO + Co(II) indicate the fact that the presence of Co(II) [60–62]. Fig. 6F shows the high-resolution N 1s spectrum of BPCAO + MB, the peaks at 395.7 eV, 399.9 eV, 402.0 eV and 405.1 eV are belonged to the positive $\equiv\text{N}^+$ combined with the negative $-\text{O}\equiv$ ($\equiv\text{N}^+ \dots -\text{O}\equiv$), N–(C)₃ [45], N=C [63, 64] and heterocycle N–C [45] in the MB adsorbed BPCAO (BPCAO + MB), respectively.

From the above analysis of BET, FT-IR and XPS results, we can conclude that in the mono-component adsorption system, the adsorption mechanism of MB on BPCAO is mainly simultaneously ruled by the multiple impacts of electrostatic interaction, π - π stacking and hydrogen bonding, resulting in an inner-sphere adsorption. Whereas, the adsorption mechanism of Co(II) on BPCAO may be controlled by the interactions of complexation or/and ion exchange. As for the MB + Co(II) simultaneous removal system, besides the mentioned interactions above, the significant interaction between MB and Co(II) can be possibly described as following: (1) the N- or S-containing groups in the MB molecule can support as complexation sites for Co(II), and enhance the adsorption of Co(II); the positive charge of Co(II) can act as an inhibited compound for positively charged MB adsorption and thus may result in a weakened sorption of MB. These influences finally cause unusual results of the simultaneous removal system at different concentrations of each sorbate.

4. Conclusions

In summary, hierarchically porous carbon (BPCA) and its oxide (BPCAO, surface functionalization with HNO_3) were successfully fabricated using banana peels as the low-cost precursor via a simple alkaline etching method. The as-prepared BPCA and BPCAO both possessed high specific surface areas, typical porous structure and excellent contaminant adsorption performances. In the

mono-component system of BPCAO, the sorption of MB and Co(II) both were well described by the pseudo-second-order and Langmuir models. The calculated thermodynamic parameters indicated that the adsorption of MB and Co(II) were spontaneous and exothermal processes. In the binary system (MB + Co(II)), the adsorption capacities of MB and Co(II) were almost unchanged at low Co(II) concentrations while the adsorption of MB was inhibited and the adsorption of Co(II) was oppositely enhanced at high Co(II) concentrations. Moreover, based on the results of BET, FT-IR and XPS, the adsorption mechanism of MB involved electrostatic interaction, π - π stacking and hydrogen bonding while that of Co(II) was mainly ruled by the complexation and/or ion-exchange. This study provided significant insights into the simultaneous removal process with the carbonaceous adsorbent and indicated that BPCAO could be a favorable multifunctional sorbent for water treatment.

Conflict of interest

The authors have no conflicts to declare.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jtice.2018.08.038.

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